

Week 5 Project Summary

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1. Planned for Week 5

Move from Week 4 planning into actual analysis: clean and verify the dataset, calculate proposed KPIs, run exploratory analysis on the 5 business questions, and build an initial dashboard.

2. What Was Completed

Completed a full data preparation review, calculated all 5 proposed KPIs, ran exploratory analysis on all 5 business questions, and built an interactive Power BI dashboard with 5 KPI cards, 5 charts, and 1 slicer (Appointment Type).

3. Key Findings

- Previous no-show history is the strongest predictor of future no-shows (43.5% to 68.8% as prior no-shows increase).
- Distance to the clinic is a moderate but real barrier (46.5% to 57.8% as distance increases).
- Reminders show only a modest effect (4-point difference).
- Waiting time and gender show no meaningful relationship with no-show behavior.

4. Major Challenges

Building the derived 'Next Appointment Outcome' calculation in Power BI (linking a patient's waiting time to their next visit's outcome) required significant troubleshooting, including a circular reference error and a column-vs-measure aggregation error that initially produced impossible percentages (over 100%). This was resolved by rebuilding the calculation as a proper DAX measure and independently verifying the result using a secondary calculation method.

5. Important Decisions Made

Grouped previous_no_shows into three bands (0, 1-2, 3+) rather than analyzing all raw values individually, since the higher values had very small sample sizes and would have produced unreliable, hard-to-read results.

6. Changes to Week 4 Approach

No major changes — the original approach (grouping patients by key factors and comparing no-show rates) was followed as planned. Waiting time analysis required an additional derived column (linking to the next appointment) that was anticipated but not yet built in Week 4.

7. Cross-Track Collaboration

Shared findings on the strongest predictors of no-shows (previous no-show history and distance) with the Data Science track, to help prioritize candidate features for their baseline prediction model.

8. Remaining Work

Further refinement of the dashboard's visual design, and deeper investigation into appointment type and age group patterns using the newly added slicers.

9. Proposed Focus for Week 6

Refine the dashboard for stakeholder presentation, explore additional segment-level patterns (appointment type, age group), and support the Data Science track with any further data clarifications needed for their modelling work.